

DAVID REY

Technical Artist | Unity & Unreal Developer

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PROFILE

Technical Artist and gameplay programmer working across Unity and Unreal Engine. Combines shaders, lighting and real-time optimization with C# and Blueprint gameplay systems. Experience includes VR environments, a published Android AR project and configurable workflows for designers. Collaborates directly with artists to turn creative requirements into efficient in-engine implementations.

TECHNICAL SKILLS

Engines and gameplay: Unity, Unreal Engine, C#, Blueprints, Animation Blueprints, ScriptableObjects, finite state machines, NavMesh and Input System.

Technical art: Shader Graph, VFX Graph, materials, lighting, lightmaps, Forward Rendering, URP, HDRP, LODs, culling and draw-call optimization.

Profiling and delivery: Unity Profiler, Frame Debugger, Unreal rendering visualization, Git, code reviews and Scrum. XR: Meta Quest, OpenXR, Vuforia and Android.

Art tools: Blender, 3ds Max, ZBrush, Substance Painter, Substance Designer and Adobe Photoshop.

EXPERIENCE

Eongate | Technical Artist and VR Developer

April 2026 - July 2026 | Remote

- Sole Technical Artist adapting Salón Dorado, a photorealistic Unreal Engine environment, for real-time VR. Investigated bottlenecks and improved scene performance from approximately 2 FPS to a stable 120 FPS.
- Replaced Lumen and dynamic lighting with Forward Rendering and baked lighting. Optimized assets with LODs, culling, material consolidation, trim sheets and reduced mesh and shader complexity.
- Guided artists transitioning from static rendering on asset preparation and performance-aware real-time workflows.

Arcadio Lab | Unity Programmer and Technical Artist

July 2024 - September 2025 | Barcelona

- Created stylized, holographic and outline effects with Shader Graph, balancing visual quality with runtime performance.
- Profiled Unity scenes with Unity Profiler and Frame Debugger to investigate rendering and runtime bottlenecks. Developed C# gameplay features and resolved issues in production builds.
- Collaborated with artists and designers in Scrum teams and participated in code reviews. Refactored external API data flows to remove unnecessary requests and avoid a higher service plan.

Forged Horizon | Unreal Engine VR Developer and Technical Artist

September 2023 - June 2024 | Barcelona

- Implemented climbing, physical doors, archery, sword combat and hand-animation behavior with Blueprints and Animation Blueprints for Golden Hand in a five-person team.
- Optimized baked lighting and post-processing for VR. The project won Deià's Best Project of the Year in 2024.

EXPERIENCE - CONTINUED

Grupo 3 Studio | Unity Programmer and Technical Artist

September 2022 - July 2023 | Barcelona

- Sole programmer in a six-person team. Created configurable mission and dialogue systems with ScriptableObjects so designers could edit content without programming.
- Implemented a C# player controller, state-machine-based animation and enemy behavior. Profiled scenes and worked with artists and designers to improve runtime stability and workflows.

SELECTED PROJECTS

I.P.O.D. Atrapamonstres | Published Android AR project | 2025

- Developed and integrated Unity gameplay interactions using Vuforia image recognition. Created capture effects, VFX and UI feedback to communicate player actions.
- Integrated external API communication and Unity Cloud services for player scores and leaderboards; reduced unnecessary external requests through data-access refactoring.

Conversational AI NPC | Unity, C# | 2026

- Built a VR prototype in four days integrating speech recognition, language-model services, speech synthesis and lip synchronization.
- Used ScriptableObjects for character configuration and decoupled, event-driven services for interchangeable local/API implementations. Added a conversation state machine to prevent invalid concurrent actions.

Game Transactions Backend | NestJS, TypeScript, PostgreSQL | 2026

- Developed a REST backend for purchases and inventory, with JWT authentication, atomic and idempotent fulfilment and automated tests. Documented API contracts and client/backend transaction flows.

EDUCATION

IT Academy - Barcelona Activa

Specialization Course in Java Spring Boot Development | September 2025 - March 2026

Deià Escola d'Art i Disseny

Audiovisual Graphics - Virtual Reality and Gamification | 2023 - 2024

Audiovisual Graphics - Video Games and Virtual Environments | 2020 - 2023

LANGUAGES

Spanish: Native | Catalan: Native | English: Advanced